Grade 4 | Computer Science

Student reading and activity booklet

36 introductory chapters aligned to the school year. Week 1 is an optional introduction; the updated schedule begins in Week 2 on October 19, 2026. Every numbered chapter occupies one PDF page, so Chapter N starts on PDF page N + 1.

Use Monday/Wednesday for Art and Life Sciences; Tuesday/Thursday for Computer Science and Life Skills. Read and model in the first session; apply, revise, and demonstrate in the second. Adapt the pace to the child. This is an original introductory booklet and practical workbook, not a claim of a complete specialist textbook.

Grade 4: explain with concrete examples, labeled drawings, and a short paragraph. Grade 7: explain using precise vocabulary, two reasons or pieces of evidence, a counterexample or limitation, and a developed paragraph.

Use fictional situations and public information. Do not require disclosure of personal beliefs, health, sexual experience, addresses, passwords, or family finances. Adults supervise tools, heat, cleaning products, health discussions, and first-aid practice. Religion is studied descriptively, without devotional participation or grading belief.

Reference and extension sources

https://www.csunplugged.org/en/

https://scratchfoundation.org/learn/learning-library

https://www.ftc.gov/media/79885

Chapter 1 | Hardware and input/output

Module 1 | Week 1 | PDF page 2

Read and discuss

Hardware is physical equipment; software supplies instructions. Input enters a system, processing changes information, storage keeps it, and output presents results.

First session: guided practice

Label computer parts and trace the path of a sentence from keyboard to saved document.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

How do input and output differ?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 2 | Typing and keyboard skills

Module 1 | Week 2 | PDF page 3

Read and discuss

Accurate typing improves through short regular practice. Locate Shift, Enter, Backspace, and space. Comfortable posture and breaks matter more than racing a timer.

First session: guided practice

Use a paper keyboard or device to practice a short paragraph twice and compare accuracy.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What is a useful typing goal?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 3 | Mouse, windows, and accessibility

Read and discuss

Selecting, scrolling, and changing windows are interface skills. Zoom, captions, and keyboard navigation provide different ways to use a system.

First session: guided practice

Complete a simple task with a pointing device and an alternative input or accessibility feature.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why provide accessibility features?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 4 | Files and folders

Module 1 | Week 4 | PDF page 5

Read and discuss

A file stores information; folders organize files. Clear names and predictable locations make work easier to find. Save copies before major changes and check where downloads go.

First session: guided practice

Create a practice subject folder; save, rename, and retrieve a document.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why use descriptive filenames?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 5 | Passwords and privacy

Module 2 | Week 1 | PDF page 6

Read and discuss

Addresses, identifying details, and passwords can be private information. Strong unique credentials and appropriate account protections reduce risk. Never put real passwords in a worksheet.

First session: guided practice

Sort fictional information by privacy needs and design a pretend passphrase.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Should a real password go on a worksheet?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 6 | Phishing and verification

Module 2 | Week 2 | PDF page 7

Read and discuss

Messages can impersonate trusted people. Urgency and unexpected requests for money or secrets are warning signs. Verify through a known route rather than following an unexpected link.

First session: guided practice

Mark warning signs in invented messages and demonstrate a safe verification plan.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Does a familiar logo prove authenticity?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 7 | Digital kindness and boundaries

Module 2 | Week 3 | PDF page 8

Read and discuss

Online actions affect real people. Ask before sharing another person's information or image. Stop interacting with unsafe content and seek trusted-adult support.

First session: guided practice

Rewrite an unkind fictional message and practice declining an unauthorized photo share.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What should follow an unsafe interaction?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 8 | Search and source evaluation

Module 2 | Week 4 | PDF page 9

Read and discuss

Search ranking does not guarantee truth. Check authorship, evidence, date, and purpose. Compare important claims with other reliable sources.

First session: guided practice

Write a focused query and evaluate two parent-selected sources.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Does appearing first in search establish accuracy?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 9 | Algorithms and precision

Module 3 | Week 1 | PDF page 10

Read and discuss

An algorithm is an ordered procedure for a task. Instructions must make order and choices clear. Testing with another person reveals missing or ambiguous steps.

First session: guided practice

Write paper-shape arrangement instructions and ask a partner to follow them literally.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why test an algorithm?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 10 | Decomposition and patterns

Module 3 | Week 2 | PDF page 11

Read and discuss

Decomposition divides a large problem into smaller tasks. Patterns can suggest reusable solutions. The plan must explain how the parts fit together.

First session: guided practice

Break a simple game into input, rules, display, and ending; show task dependencies.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What is decomposition?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 11 | Binary information

Module 3 | Week 3 | PDF page 12

Read and discuss

A bit has two possible values, commonly written 0 and 1. Binary places represent powers of two. Patterns of bits can encode information.

First session: guided practice

Use cards labeled 1, 2, 4, 8, and 16 to encode and decode small numbers.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What values can a bit have?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 12 | Pixels and images

Module 3 | Week 4 | PDF page 13

Read and discuss

A digital image can be represented as a grid of pixels holding color information. More pixels can support detail but do not guarantee better artistic composition.

First session: guided practice

Encode a small black-and-white grid, exchange codes, and reconstruct the image.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What is a pixel?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 13 | Scratch sequences

Module 4 | Week 1 | PDF page 14

Read and discuss

Connected Scratch blocks form scripts. Start with a small instruction sequence and test it before adding complexity. Order affects behavior.

First session: guided practice

Make a sprite move and speak, or trace equivalent paper blocks; predict a reordered sequence.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why does order matter?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 14 | Events and interaction

Module 4 | Week 2 | PDF page 15

Read and discuss

An event triggers instructions, such as a click or key press. Programs may have several event-driven scripts. State which event starts each action.

First session: guided practice

Create a key-press action or paper event-response diagram and test two inputs.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What is an event in a program?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 15 | Loops and repetition

Module 4 | Week 3 | PDF page 16

Read and discuss

A loop repeats instructions. Some loops count repetitions; others depend on a condition. Loops can express repetition clearly with less duplicate code.

First session: guided practice

Draw a square with movement-and-turn steps, then replace repeated steps with a loop.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why use a loop?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 16 | Conditions and decisions

Module 4 | Week 4 | PDF page 17

Read and discuss

A condition is evaluated as true or false. If-then rules decide when actions happen. Test cases that satisfy and do not satisfy the condition.

First session: guided practice

Create a rule for changing a sprite's appearance and test both outcomes.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why test both outcomes?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 17 | Variables and scores

Module 5 | Week 1 | PDF page 18

Read and discuss

Variables store values that can change. A score needs a known starting value and clear update rules. Test restarts as well as successful play.

First session: guided practice

Design a score counter and trace its value through a sequence of events.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why initialize a variable?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 18 | Coordinates and movement

Module 5 | Week 2 | PDF page 19

Read and discuss

Coordinates locate positions with ordered values. On a standard plane, x is horizontal and y vertical. State units and scale when they matter.

First session: guided practice

Plot sprite positions on graph paper and describe coordinate changes.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What does changing x usually represent?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 19 | Randomness and simulation

Module 5 | Week 3 | PDF page 20

Read and discuss

Simulations model selected rules of a system. Random choices represent variation but do not guarantee equal short-run outcomes. Repeat trials before comparing patterns.

First session: guided practice

Simulate a die using paper or code and record several trials.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Does a short trial guarantee an even distribution?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 20 | Debugging and tests

Module 5 | Week 4 | PDF page 21

Read and discuss

Debugging compares expected and actual behavior, changes a relevant feature, and tests again. Keep records so useful tests can be repeated.

First session: guided practice

Find a mistake in a short sequence and write a test case that would catch it.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What belongs in a debugging record?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 21 | Game goals and rules

Module 6 | Week 1 | PDF page 22

Read and discuss

Games need understandable goals, rules, and feedback. Plan winning, losing, and restarting. Fair challenge differs from an impossible task.

First session: guided practice

Write a one-page educational-game design and ask a partner to explain the rules.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why test rule clarity?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 22 | Game feedback and balance

Module 6 | Week 2 | PDF page 23

Read and discuss

Feedback communicates what happened after an action. Balance adjusts challenge and reward for an audience. Observe testers rather than assuming they behave like the designer.

First session: guided practice

Add a visible response to an action and revise one confusing feature after testing.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What is game feedback?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 23 | Interfaces and accessibility

Module 6 | Week 3 | PDF page 24

Read and discuss

Interfaces connect people and systems. Labels, readable contrast, alternative inputs, and clear feedback improve usability. Decoration should support comprehension.

First session: guided practice

Sketch two versions of game controls and compare their readability and input options.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why label controls clearly?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 24 | Copyright and attribution

Module 6 | Week 4 | PDF page 25

Read and discuss

A license specifies permissions and restrictions for reusing work. Giving credit does not automatically authorize every use. Original assets can avoid reuse restrictions.

First session: guided practice

Record the creator and license of an approved image, then create an original alternative.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Does credit always grant permission?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 25 | Tables and spreadsheets

Module 7 | Week 1 | PDF page 26

Read and discuss

Tables organize records and attributes. Spreadsheet formulas depend on correct inputs and rules. Label units and check calculated results by another method.

First session: guided practice

Enter fictional plant data and calculate a total or average; verify it by hand.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why check a spreadsheet formula?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 26 | Graphs and misleading displays

Module 7 | Week 2 | PDF page 27

Read and discuss

Graph scales, labels, and data selection affect interpretation. A numerically accurate graph may still mislead. Inspect units and axis ranges.

First session: guided practice

Graph one fictional data set using two different axis ranges and compare impressions.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why inspect an axis scale?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 27 | Networks and the internet

Module 7 | Week 3 | PDF page 28

Read and discuss

A network connects devices exchanging information. The internet is a network of networks; the web is one service using it. Information passes through several systems.

First session: guided practice

Draw a simplified route from a device to a website and explain the parts.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Are the internet and web identical?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 28 | Security and backups

Module 7 | Week 4 | PDF page 29

Read and discuss

Backups preserve recoverable copies. Updates can address security problems. Test recovery using harmless practice files and parent-approved tools.

First session: guided practice

Back up a practice file, restore it under a new name, and compare contents.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

How can a backup be checked?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 29 | Web-page structure and HTML

Module 8 | Week 1 | PDF page 30

Read and discuss

HTML marks a page's structure and meaning. Headings, paragraphs, and lists have different roles. Browsers interpret markup to display content.

First session: guided practice

Create a small local page or paper markup plan with a heading, paragraph, and list.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What is HTML for?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 30 | CSS and layout

Module 8 | Week 2 | PDF page 31

Read and discuss

CSS describes presentation, including spacing and color. Consistency helps navigation. Check whether a design remains readable at different sizes.

First session: guided practice

Style a local page or paper mockup and compare wide and narrow layouts.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

How does CSS differ from HTML?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 31 | Interactive models and mechanics

Module 8 | Week 3 | PDF page 32

Read and discuss

A model changes in response to input according to explicit rules. State units and limitations before interpreting outputs. Models simplify real systems.

First session: guided practice

Design a program or paper model of a lever or measuring tool and test its input/output rules.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why state limitations?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 32 | Project testing

Module 8 | Week 4 | PDF page 33

Read and discuss

A finished project should meet stated requirements and handle ordinary and unusual inputs. Record failures honestly and retest fixes. One success does not establish universal correctness.

First session: guided practice

Write a test checklist and ask a partner to test a project.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Does one successful run prove universal correctness?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 33 | AI and generated information

Module 9 | Week 1 | PDF page 34

Read and discuss

AI can generate plausible but inaccurate information. Confidence is not evidence. Protect private data and check important claims with reliable sources.

First session: guided practice

Compare a fictional confident answer with supplied evidence and label unsupported claims.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Does confidence establish an AI answer's accuracy?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 34 | Project plans and versions

Module 9 | Week 2 | PDF page 35

Read and discuss

Plans specify requirements, steps, and completion checks. Versioned copies support recovery and comparison. Small tested changes are easier to understand than many untested changes.

First session: guided practice

Save a starting version, implement one improvement, and compare results.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Why save versions?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 35 | Documentation

Module 9 | Week 3 | PDF page 36

Read and discuss

Documentation explains a project's purpose, inputs, outputs, operation, and limits. Write for a person who did not build it. Test whether the guide works without extra coaching.

First session: guided practice

Write a short user guide and have a tester follow only that guide.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

Who needs documentation?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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Chapter 36 | Showcase and reflection

Module 9 | Week 4 | PDF page 37

Read and discuss

Demonstrate a working feature and explain design choices with evidence. Reflection identifies strengths, limits, and next steps. Credit borrowed resources and collaborators.

First session: guided practice

Present a feature, test record, user feedback, and a justified improvement.

Second session: apply and revise

Use a concrete example and labeled drawing; explain your result in 3-5 sentences. Review the first attempt with your parent and improve one specific feature.

Check your understanding

What strengthens a project presentation?

My answer: _____

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Activity record: sketch, steps, measurements, or observations

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One revision and why it helps: _____

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